

Instructions & Defense Runbook

How to set up, verify, regenerate, and defend every number in the submission · repo:
github.com/Manureddy148/AI_Task2 · compiled 2026-09-04

1 Setup

```
git clone https://github.com/Manureddy148/AI_Task2.git && cd AI_Task2
pip install -r submission/requirements.txt          # Python 3.10+; no torch, no GPU
```

Network is needed once: tokenizer files (tiktoken/HuggingFace) and, only if you rebuild the corpus, the public FLORES-200 tarball (24 MB). **Layout:** scripts find the audited inputs via `submission/_paths.py` — the committed `starter_kit/` works as-is; any other layout: set `STARTER_KIT=<path containing bench/>`.

2 Verify first (the 30-second credibility check)

```
python submission/verify.py          # expected final line: ALL 50 NUMBERS VERIFIED
python submission/verify.py --full   # also regenerates results/romanization/reconcile first (~3 min)
```

The 50 assertions cover: byte-exact REPORT_v0 reproduction, each bug's isolated delta, corpus SHA-256 pins, the full parity grid, every derived ratio quoted in prose, romanization on both tokenizer generations, all 13 bench-log reconciliations, and the roofline fit. CI runs the same harness on every push.

Defense preparation: `python submission/defense_drill.py` walks the candidate's personal checklist interactively (~10 min) — runs verify, the A0 repro, a live corpus toggle, B1 on paper, and one bench row by hand, showing expected vs actual at each step.

3 Regenerate everything from scratch

```
python submission/partA/corpus/build_corpus.py      # corpus (downloads on first run; hash-pinned)
python submission/partA/audit/experiments.py        # A2 one-toggle bug isolation
python submission/partA/fertility_v1.py              # A3 grid: 5 tokenizers × 4 denominators × 7 languages
python submission/partA/audit/romanization.py        # romanization + Tamil artifact bound + probes
python submission/partB/kv_math.py                  # B1 arithmetic
python submission/partB/reconcile.py                 # 13-row replay, goodput, roofline, fp8 forecast
python submission/figures/make_figures.py            # report figures (annotations computed from data)
python submission/verify.py                          # assert it all again
```

4 Live-defense overrides ("change this input right now")

Ask	Command
Paste a modified toy corpus into A2	<code>python submission/partA/audit/experiments.py --hin pasted.txt</code> — every variant, the data facts and the bootstrap CI move with it (sizes are derived, not hardcoded).
Swap one language's file in A3	<code>python submission/partA/fertility_v1.py --corpus hin=pasted.txt</code> (repeatable; line-alignment with eng is asserted with a clear error).

Point Part B at an edited bench log	<code>python submission/partB/reconcile.py --log edited.csv</code> — all seven sections recompute; row binding is unique-match asserted.
Change a spec constant	Edit the constant in <code>kv_math.py</code> — every printed equation string is an f-string over the constants; nothing numeric is retyped literal text.
Different starter-kit location	<code>STARTER_KIT=<path></code> env var.

5 Defense crib — the numbers and where each one lives

Number	Derivation	Source of truth
114,688 B/token KV (112 KiB)	$2 (K,V) \times 28 \text{ layers} \times 8 \text{ KV heads} \times 128 \text{ head_dim} \times 2 \text{ B fp16}$	partB/kv_math.py
25.7 → ~25 concurrent 4k seqs	$(0.92 \cdot 24 - 8.4 - 1.6) \text{ GB} \div (4096 \times 114,688 \text{ B})$	kv_math.py; log confirms: util 0.93@24, preempt 7=32–25, 23=48–25
201 tok/s honest goodput	512·24/61.16 by definition; independent route n/itl = 249.8 tok/s over the 49.1 s decode phase; counter/8 is an identity, not evidence	reconcile.py §3; capacity.md B3
65% MBU roofline	$\text{itl} = (8.4 \text{ GB} + \text{resident_avg_ctx} \cdot 112 \text{ KiB}) / (0.65 \cdot 300 \text{ GB/s})$ fits 13/13 ITLs $\pm 2.8\%$; compute 1.7 ms vs 63.2 ms memory → memory-bound ~38×	reconcile.py §6
Parity: hin 7.42× · kan 13.59× · tam 15.54× · tel 12.97× · ben 9.61× · mar 7.86×	total tokens ÷ English tokens over the same 1012 parallel sentences (gpt2 = v0's tokenizer); CIs $\leq \pm 1.28\%$	partA/results/results.csv; fig1
MuRIL 1.00–1.20×	Same corpus, Indic-focused vocab; unk $\leq 0.38\%$ (integrity column)	results.csv
Romanized hin 2.25× (gpt2) / 1.87× (o200k)	IAST-derived chat approximation; organic-convention probe 2.11/1.76; Tamil is a range 2.37–3.09× (aspirate artifact bound)	partA/audit/romanization_output.md
Bug deltas +0.59 / +2.92 / +0.35%	one-toggle isolation vs byte-exact v0 baseline 5.8871×	partA/audit/experiment_output.md

6 The five hardest defense questions — and where the answer is

- "Why is tokens-per-word wrong? It's the standard fertility metric." Fine *within* a language; across languages it errs –15% (hin) to +36% (kan) against content-constant truth because words carry different content per language — findings.md C1 table, derived from results.csv.
- "Your two metrics agreed — doesn't that confirm the result?" Shared numerator ⇒ correlation by construction; the repo shows the same tokens giving 6.34× (word) / 11.39× (grapheme) / 2.90× (byte) / 7.46× (code point) — findings.md C2 plus results.md's denominator section. (And the audit applied this to itself: see B3's relabelled "identity".)

- **"Is 7.4× what production pays?"** Unknown — the spec's 128k vocab is not gpt2; the premium is bracketed (cl100k row is scale-illustration) until recommendation 6 measures the real tokenizer. REPORT_v1 §4.
- **"Why does throughput fall at batch 32?"** 448 MiB/seq × 32 > 12.08 GB budget → 7 evictions with full re-prefill (~3.5 s trigger) + serialized tail past the 25-slot ceiling; util pegs at 0.97 — capacity.md B2, every column reconciled.
- **"What would falsify your fixes?"** Stated in advance: max_num_seqs=24 → 122.3 s / 0 preemptions at offered 48; fp8-KV → 51 slots, ~2× decode (capacity half is arithmetic-certain, ITL half assumes MBU transfers); counter to pull: vllm:num_preemptions_total ≥7/≥23 — capacity.md B2/B4.

7 Troubleshooting

- **HF rate-limit warning** on first tokenizer download is benign; set HF_TOKEN to silence.
- **Corpus hash FAIL after a fresh clone** would mean newline translation — the repo ships .gitattributes *-text precisely to prevent it; check that file survived any re-zip.
- **verify.py --full failure**: the failing child's stderr prints through (stdout is suppressed); fix the cause it names.
- Windows/macOS/Linux all fine; everything is CPU-only.

Regenerate these PDFs: powershell docs/make_pdfs.ps1 (headless Edge/Chromium print of docs/src/*.html).